

Unit-I : Quantum Computing - MCQs

1. In quantum computing, what is the basic unit of information?

- a) Giga
- b) Qubit
- c) Bit
- d) Byte

Answer: Qubit

2. What do we call the pieces of information in a quantum computer?

- a) Bits
- b) Qubits
- c) Bytes
- d) Qubytes

Answer: B

3. When the information is between 0 and 1 in a quantum computer, what do we call this?

- a) Superposition
- b) Same position
- c) Ordinary position
- d) Different position

Answer: A

4. Quantum computers are very good at dealing with_____

- a) Clarity
- b) Certainty
- c) Uncertainty
- d) Reliability

Answer: C

5. What does 'entanglement' mean?

- a) Two particles are different
- b) Two particles are separate
- c) Two particles are independent
- d) Two particles are connected

Answer: D

6. What can quantum computers be used for?

- a) Artificial Intelligence
- b) Simulations/Predictions
- c) Both (A) and (B)
- d) Google Docs and Slides

Answer: C

7. When the two members of a Qubit pair exist in a single quantum state, it is known as _____.

- a) Entanglement
- b) Engagement

- c) Superposition
- d) None of the Above

Answer: A

8. Quantum computing is relatively _____ than classical computing.
- a) Faster
 - b) Slower
 - c) Average
 - d) None of the Above

Answer: A

9. Qubit stands for _____
- a) Quality bits
 - b) Question bit
 - c) Quantum gates
 - d) Quantum bit

Answer: D

10. A qubit is a _____ quantum-mechanical system.
- a) One-state
 - b) Two-state
 - c) Three-state
 - d) Four-state

Answer: B

11. The set of vectors and set of scalars which follow the same properties followed by linear vector space is said be
- a. Basis
 - b. Dimension
 - c. Hilbert space
 - d. Orthogonal state

Answer: C

12. It is the process of replacing i^{th} row of the matrix by i^{th} column, then it is said to be
- a. Conjugate Matrix
 - b. Transpose Matrix
 - c. Identity Matrix
 - d. Hermitian Operator

Answer: B

13. The operators change with time while the state vectors remain constant, then it is said to be
- a. Schrodinger representation
 - b. Heisenberg representation
 - c. Interaction representation
 - d. None of the above

Answer : B

14. The operators remain constant while the state vectors change with time, then it is said to be
- Schrodinger representation
 - Heisenberg representation
 - Interaction representation
 - None of the above

Answer : A

15. The diagonal entries of a Hermitian matrix must be
- Complex conjugate
 - Real
 - Both real & Complex conjugate
 - None of the above

Answer : B

16. The eigen value of a Hermitian matrix must be
- Complex conjugate
 - Real
 - Both real & Complex conjugate
 - None of the above

Answer : B

17. What is a vector space?
- A space consisting of only vectors
 - A set of vectors closed under addition and scalar multiplication
 - A space that includes both vectors and scalars
 - A space that is always three-dimensional

Answer: b. A set of vectors closed under addition and scalar multiplication

18. What is the dimension of a vector space?
- The size or length of a vector
 - The number of vectors in the space
 - The maximum number of linearly independent vectors that span the space
 - The number of elements in the basis of the space

Answer: c. The maximum number of linearly independent vectors that span the space

19. What is the span of a set of vectors?
- The set of all vectors in the vector space
 - The linear combination of all vectors in the set
 - The set of vectors that are orthogonal to the given set
 - The set of vectors that are linearly independent

Answer: b. The linear combination of all vectors in the set

20. In a finite-dimensional vector space, what is the maximum number of linearly independent vectors a basis can have?
- a. 0
 - b. 1
 - c. The dimension of the vector space
 - d. The size of the vector space

Answer: c. The dimension of the vector space

21. Moore's Law originally stated that the number of transistors on a microchip would double approximately every:
- a. 6 months
 - b. 1 year
 - c. 2 years
 - d. 5 years

Answer: c. 2 years

22. What fundamental technology trend enabled the continuation of Moore's Law for several decades?
- a. Miniaturization of transistors
 - b. Increase in clock speed
 - c. Expansion of data storage
 - d. Advancements in software algorithms

Answer: a. Miniaturization of transistors

23. Which component of a computer is primarily affected by Moore's Law?
- a. Central Processing Unit (CPU)
 - b. Random Access Memory (RAM)
 - c. Hard Disk Drive (HDD)
 - d. Graphics Processing Unit (GPU)

Answer: a. Central Processing Unit (CPU)

24. What is one of the main factors contributing to the end of Moore's Law?
- a. Decreased demand for computing power
 - b. Physical limits of miniaturization
 - c. Lack of innovation in software development
 - d. Increasing costs of semiconductor production

Answer: b. Physical limits of miniaturization

25. Which alternative approaches are being explored to extend computing power beyond the limits of Moore's Law?
- a. Quantum computing
 - b. Neuromorphic computing
 - c. Optical computing
 - d. All of these

Answer: d. All of these

26. What is the fundamental unit of information in quantum computing?

- a. Bit
- b. Byte
- c. Qubit
- d. Quantum gate

Answer: c. Qubit

27. In classical computing, information is processed using bits. What are the two possible values for a classical bit?

- a. 0 and 1
- b. True and False
- c. -1 and 1
- d. Red and Blue

Answer: a. 0 and 1

28. Which property allows qubits to represent multiple states simultaneously in quantum computing?

- a. Superposition
- b. Entanglement
- c. Interference
- d. Tunnelling

Answer: a. Superposition

29. In a CNOT gate, you create a(n) _____ with two qubits.

- a. Superposition
- b. Entangled state
- c. Bloch
- d. Hadamard

Answer: b. Entangled state

30. In a quantum circuit, this gate is used to place a qubit into superposition.

- a. Hadamard
- b. X-gate
- c. Bloch
- d. CNOT

Answer: a. Hadamard

31. This quantum gate acts on a single qubit and would most be similar to a traditional NOT gate.

- a. CNOT
- b. X-Gate
- c. Hadamard
- d. Deutsch Gate

Answer:

